

Tutorial 05

CS105.3 Database Management Systems

Question 01

Design a database to keep track of information for Computer Services Company. Assume that the following requirements were collected. Draw an EER schema diagram for this application. Discuss any assumptions you make.

A Computer Services Company needs a system for the payment of consultants and billing of clients. The company manages IT development and technical support services such as the procurement, installation, and maintenance of ICT systems for a range of customers who are billed monthly. They have a pool of freelance consultants they call upon to do the work. Currently they keep a set of index cards listing the consultants' contact details and their skills set (i.e. which analysis and design methodologies, programming languages, operating systems, communications technologies, hardware platforms and database packages they are familiar with etc.).

A passport photo of the consultant is stapled onto the index card. Similarly, they maintain customer files with the customer's primary contact details, the billing address and copies of the invoices. Each invoice has a tick-box to indicate if it has been paid. The customer may have a number of sites where the work takes place. Each job is assigned a project/budget number by the company and the site, and a brief description of the work recorded.

The free-lance consultants submit weekly worksheets listing the dates, hours, type of work and budget code of the projects they have been working on so the company can pay them and invoice the clients. Consultants are paid by an hourly rate. There are different rates for the different types of work: strategic consultancy (including systems analysis and design), system development and implementation, and technical support. On each project, in addition to their normal work, one of the consultants is given responsibility as account manager for which they get paid an extra 50% on the normal rate.